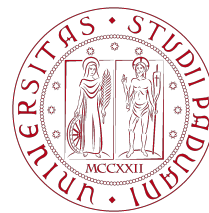


Final Report

Physics of Complex Networks: Structure and Dynamics



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Areas of physics by complexity



Newton's
Mechanics

Electro-
Magnetism

Special
Relativity

Quantum Mechanics
General Relativity

Quantum
Field Theory

Complexity
Science

Project # X:

Surname, name

Last update: July 5, 2026

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1 | Epidemic spreading on temporal networks

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1.1 | Models and mean-field thresholds

We study the four canonical compartmental models on *time-varying* contact networks. A node is Susceptible (S), Exposed (E, infected but not yet infectious), Infectious (I) or Removed (R). In discrete time, on every contact of snapshot t an S–I pair transmits with probability β ; an infectious node leaves state I with probability μ , and in SEIR an exposed node matures with probability σ . The four models are the same core loop restricted to different states [2]: SI (no recovery), SIS (I→S), SIR (I→R) and SEIR (S→E→I→R).

The order parameter is the endemic prevalence (SI/SIS) or the final epidemic size (SIR/SEIR), which is non-zero only above a critical ratio $(\beta/\mu)_c$. On the *activity-driven* temporal network the heterogeneous mean-field threshold is governed by the *activity* moments and not by the aggregated degree [3],

$$\left(\frac{\beta}{\mu}\right)_c^{\text{AD}} = \frac{1}{m(\langle a \rangle + \sqrt{\langle a^2 \rangle})}, \quad (1.1)$$

whereas treating the whole observation window as a single *static* graph gives the classical heterogeneous- (HMF) and quenched- (QMF) mean-field results evaluated on that aggregate,

$$\left(\frac{\beta}{\mu}\right)_c^{\text{HMF}} = \frac{\langle k \rangle}{\langle k^2 \rangle}, \quad \left(\frac{\beta}{\mu}\right)_c^{\text{QMF}} = \frac{1}{\lambda_{\max}(A)}. \quad (1.2)$$

Because the aggregate accumulates every contact ever made, $\langle k^2 \rangle$ and $\lambda_{\max}$ are large and the static threshold is orders of magnitude below Eq. (1.1): assuming that contacts are simultaneous systematically over-estimates spreading. Making this contrast explicit is the aim of the task.¹

1.2 | Synthetic temporal topologies and method

A temporal network is a sequence of T single-step snapshots. We use two structurally different synthetic generators, so the models are exercised across a variety of topologies.

¹The simulator uses discrete-time per-step probabilities (β, μ) , which map onto the continuous rates of Eqs. (1.1)–(1.2) for $\beta, \mu \ll 1$; the agreement in Fig. 1.1(b) confirms the mapping in the regime studied.

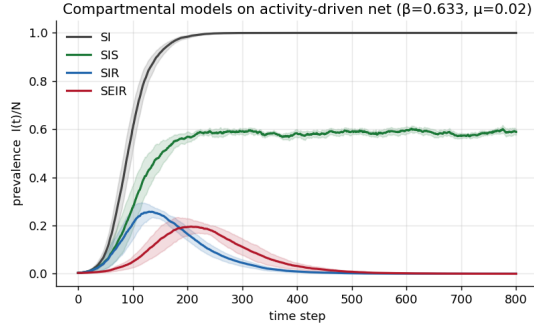
(i) The *activity-driven* (AD) model [3]: each node draws an activity $a_i \propto x^{-\gamma}$, $x \in [\varepsilon, 1]$; at every step it is active with probability a_i and, if so, fires m links to random partners. There is no fixed backbone. (ii) An *edge-activated* process on a fixed Barabási–Albert (or Erdős–Rényi) backbone whose edges switch on independently with probability p each step, so the aggregate is the backbone itself. Against each temporal run we compare two null models built from the same data: the *aggregated static* network (all edges present at every step) and a *time-shuffled* surrogate (contacts reassigned to random steps, destroying temporal order while preserving the aggregate and the total contact volume). A single seeded stochastic engine (Python + `igraph`) runs all four models on any of these sequences; results are averaged over 15 realisations. Default: $N = 1500$, $T = 800$, $m = 1$, $\gamma = 2.1$, $\mu = 0.02$.

1.3 | Results

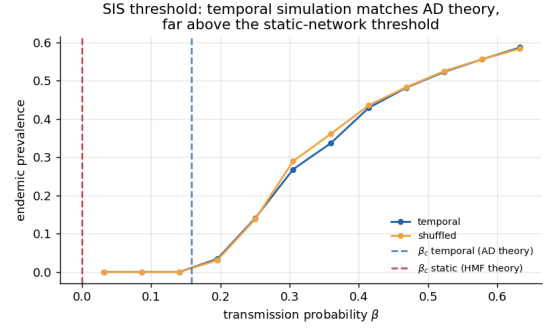
Figure 1.1(a) shows the four models on one AD network above threshold: SI saturates, SIS relaxes to an endemic plateau, and SIR/SEIR are single outbreaks, with SEIR delayed by the latent stage. Panel (b) locates the SIS threshold: the simulated prevalence departs from zero exactly at the activity-driven prediction β_c^{AD} of Eq. (1.1) (Table 1.1), far above the static-network value β_c^{HMF} . The time-shuffled curve coincides with the temporal one, confirming that in a memoryless activity-driven model the *ordering* of contacts is immaterial – the relevant effect is concurrency, not order. Panels (c,d) isolate that concurrency effect for SIR: collapsing the timeline into the aggregated static graph turns a slow, bounded outbreak (peak prevalence ≈ 0.18) into an explosive one (peak ≈ 0.98 , final size 1), on both the activity-driven and the edge-activated topologies. Treating a temporal network as static is therefore not a harmless approximation: it moves the epidemic threshold by nearly three orders of magnitude and grossly inflates the predicted outbreak.

	β_c^{AD} (temporal)	β_c^{HMF} (static)	β_c^{QMF} (static)
$(\beta/\mu)_c$	7.91	9.5×10^{-3}	1.1×10^{-2}
β_c (at $\mu = 0.02$)	0.158	1.9×10^{-4}	2.2×10^{-4}

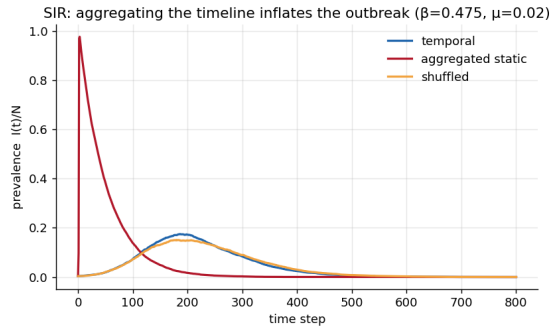
Table 1.1: Critical points for the default AD network ($\langle a \rangle = 0.041$, $\langle a^2 \rangle = 7.4 \times 10^{-3}$; aggregate $\langle k \rangle = 61$, $\langle k^2 \rangle = 6.4 \times 10^3$). The temporal threshold is $\sim 800\times$ the static one.



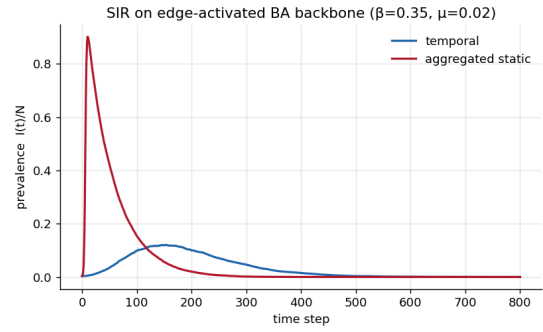
(a) Four models on the AD network



(b) SIS threshold vs. theory



(c) SIR: temporal vs. aggregated (AD)



(d) SIR on the edge-activated BA backbone

Figure 1.1: Epidemics on temporal networks. (a) SI/SIS/SIR/SEIR prevalence on one activity-driven network ($\beta = 4\beta_c^{\text{AD}}$); bands are $\pm 1\sigma$ over realisations. (b) SIS endemic prevalence vs. β : the temporal simulation matches β_c^{AD} (dashed) and lies far above β_c^{HMF} ; time-shuffled $\approx$ temporal. (c,d) Aggregating the timeline (red) inflates the SIR outbreak relative to the temporal dynamics (blue) on both topologies.

Code and data (generators, engine, thresholds, edge lists) are in `code/task_22/` and `data/task_22/`; see also Holme & Saramäki [1] for the temporal-network framework.

2 | Bibliography

- [1] Petter Holme and Jari Saramäki. Temporal networks. *Physics Reports*, 519(3): 97–125, 2012.
- [2] Romualdo Pastor-Satorras, Claudio Castellano, Piet Van Mieghem, and Alessandro Vespignani. Epidemic processes in complex networks. *Reviews of Modern Physics*, 87(3):925–979, 2015.
- [3] Nicola Perra, Bruno Gonçalves, Romualdo Pastor-Satorras, and Alessandro Vespignani. Activity driven modeling of time varying networks. *Scientific Reports*, 2:469, 2012.